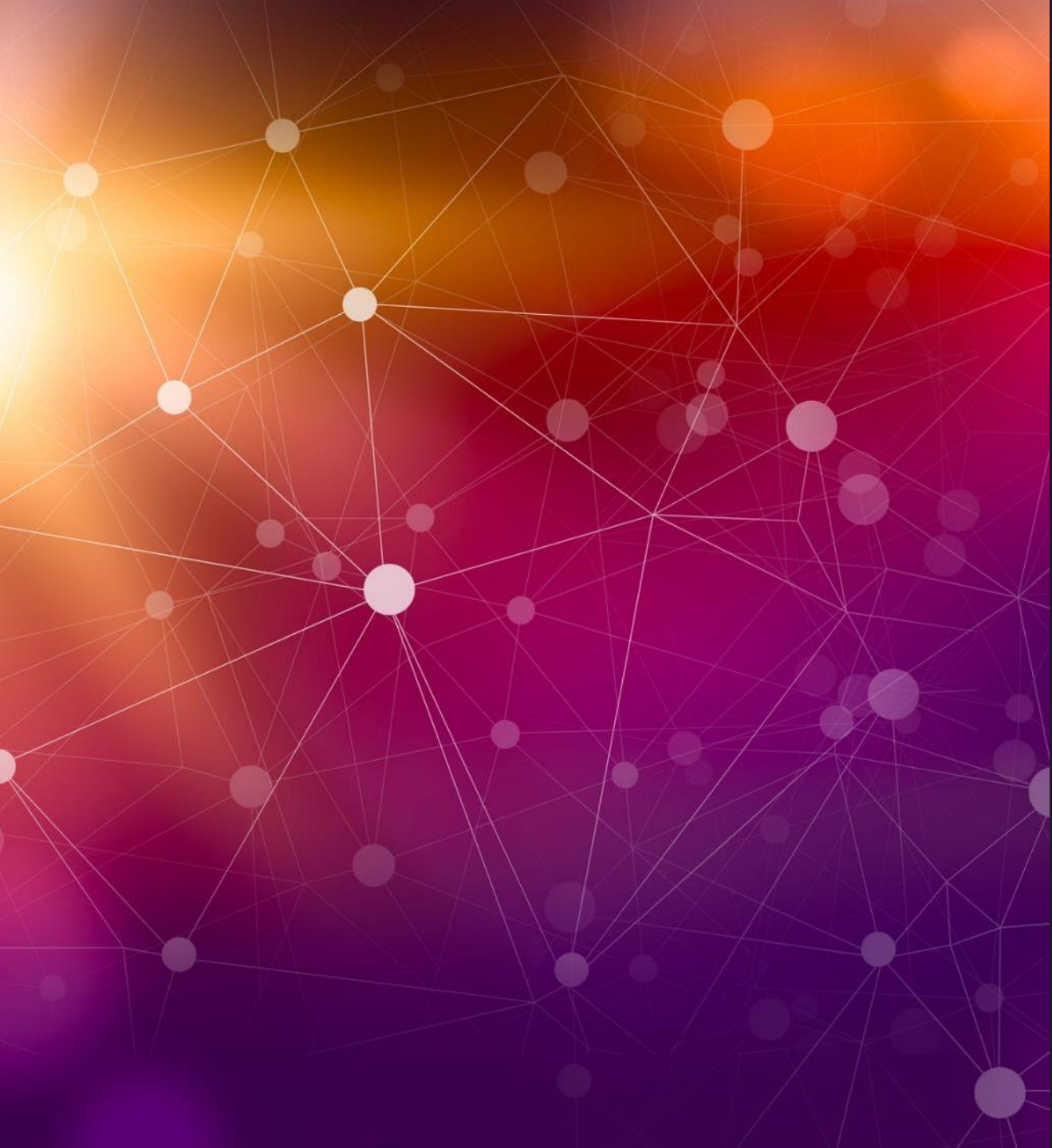




Blizzard x IonQ

Ismael Gonzalez
Shane Tendo
Lukas Rapp
Félix Wilhelmy
Aaron Kim

Outline



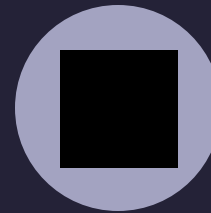
OVERVIEW OF THE
PROBLEM
STATEMENT



OVERVIEW OF OUR
FINAL RESULT



GAME THEORY



CIRCUIT THEORY



RESULTS

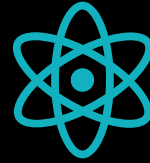
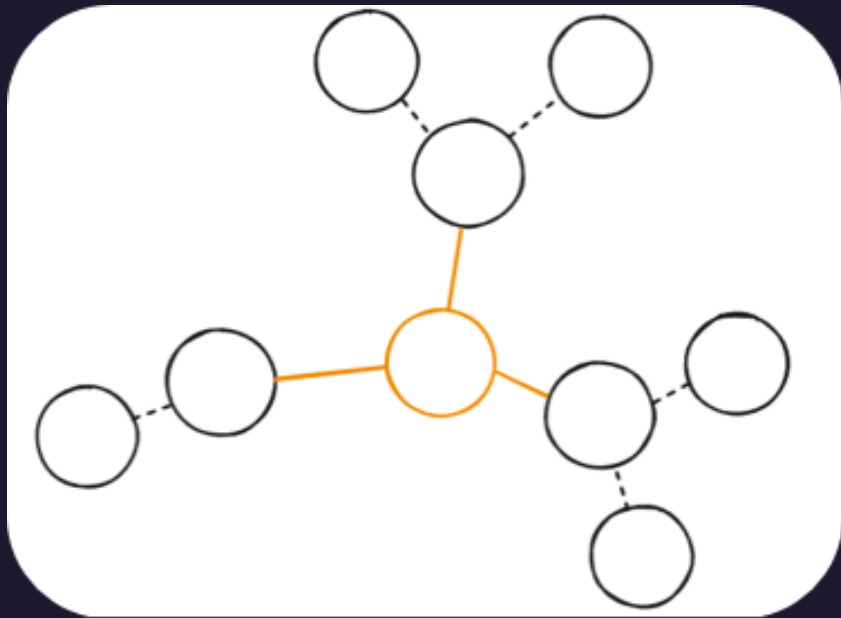


FUTURE
IMPROVEMENTS



Problem Statement

Objective: Build a quantum network subgraph to maximize your score



Quantum: Design distillation circuits



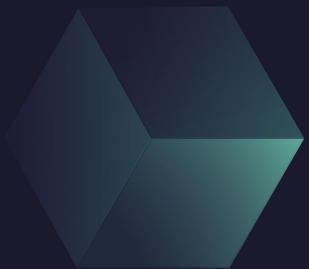
Classical: Optimize attack strategy



Economics: Manage Bell pair budget

Circuit Theory

The strategy will be multiple pre-made circuits tailor-made for maximum impact, all divided along the middle



Intro to distillation

With distillation, we can turn spread an entangled state out over multiple qubits and use that

Noise Model and Distillation Strategie

- **First Step:** understanding the noise
- **Idea:** use distillation circuit to estimate noise type
- **Insights:**
 - 1) Noise has either only X or Z errors
 - 2) Odd/ Even Difficulty has X/ Z errors

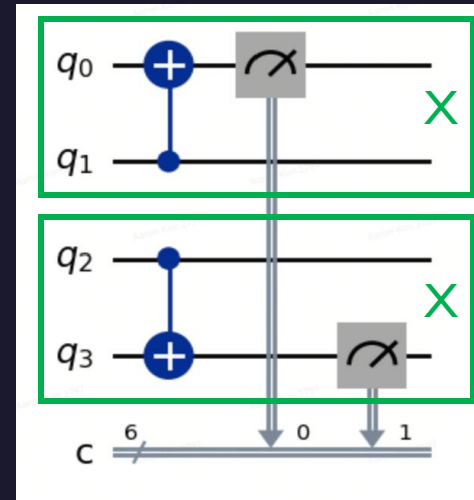
Difficulty	X Errors	Z Errors
1	15 %	0 %
2	0 %	20 %
3	25 %	0 %
4	0 %	28 %
5	?	?



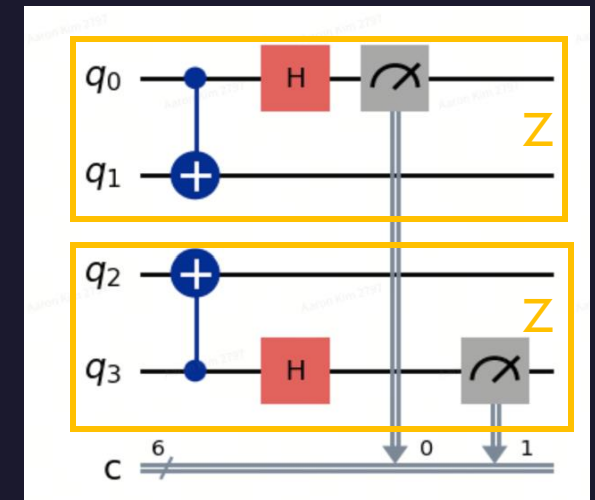
Distillation: Basic Building Blocks

- **Challenge:** local quantum + global classical operations
- **Idea:**
- CNot measures bell pair parities locally.
- Bit logic check detects incorrect pairs discarded via **flag bit**

Distillation with X Errors

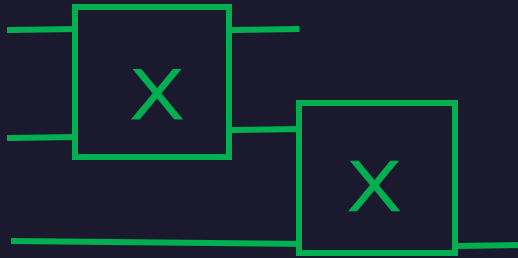


Distillation with Z Errors

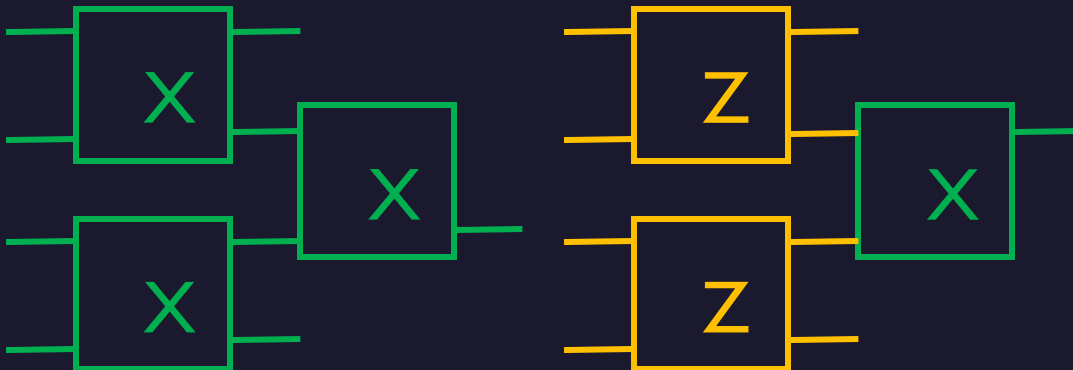


Modular Distillation

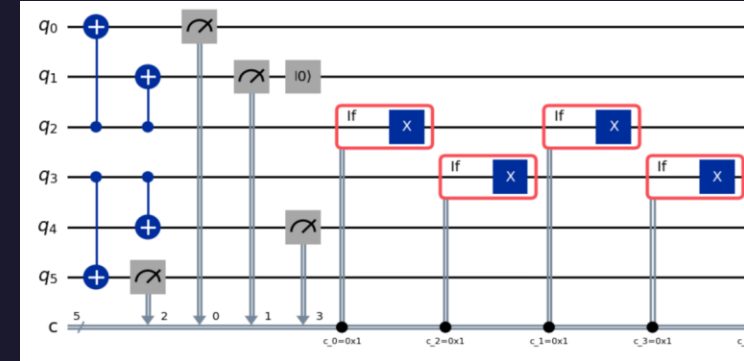
3-Qubit Pair



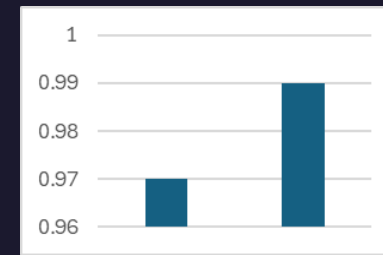
4-Qubit Pair



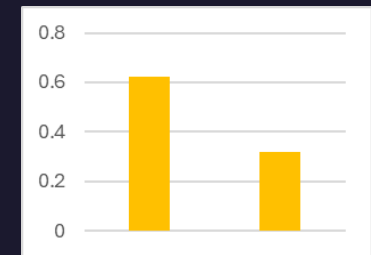
Beyond: 3 Bell Pair Circuit



Circuit Performance



Fidelity

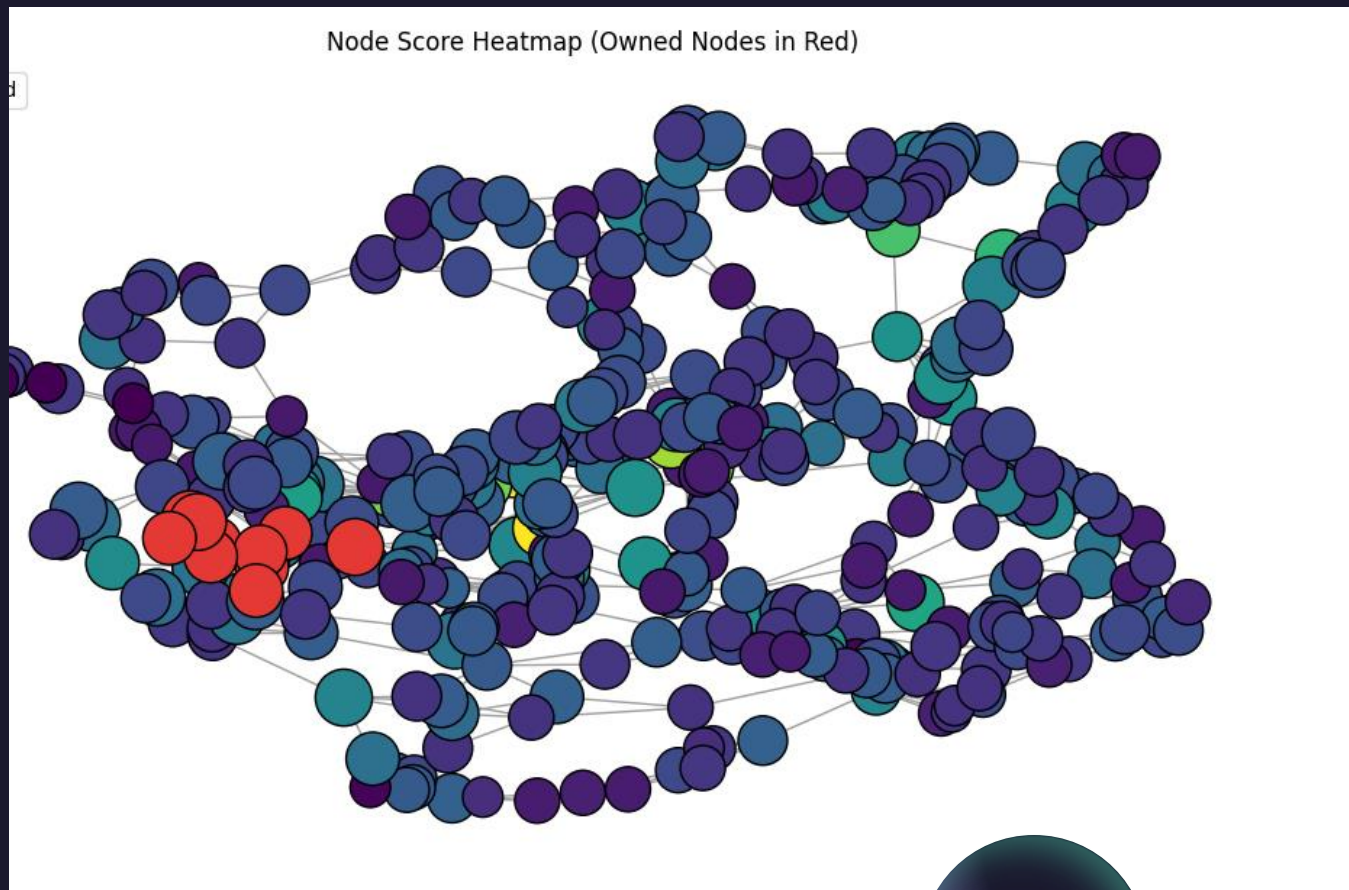


Success Rate



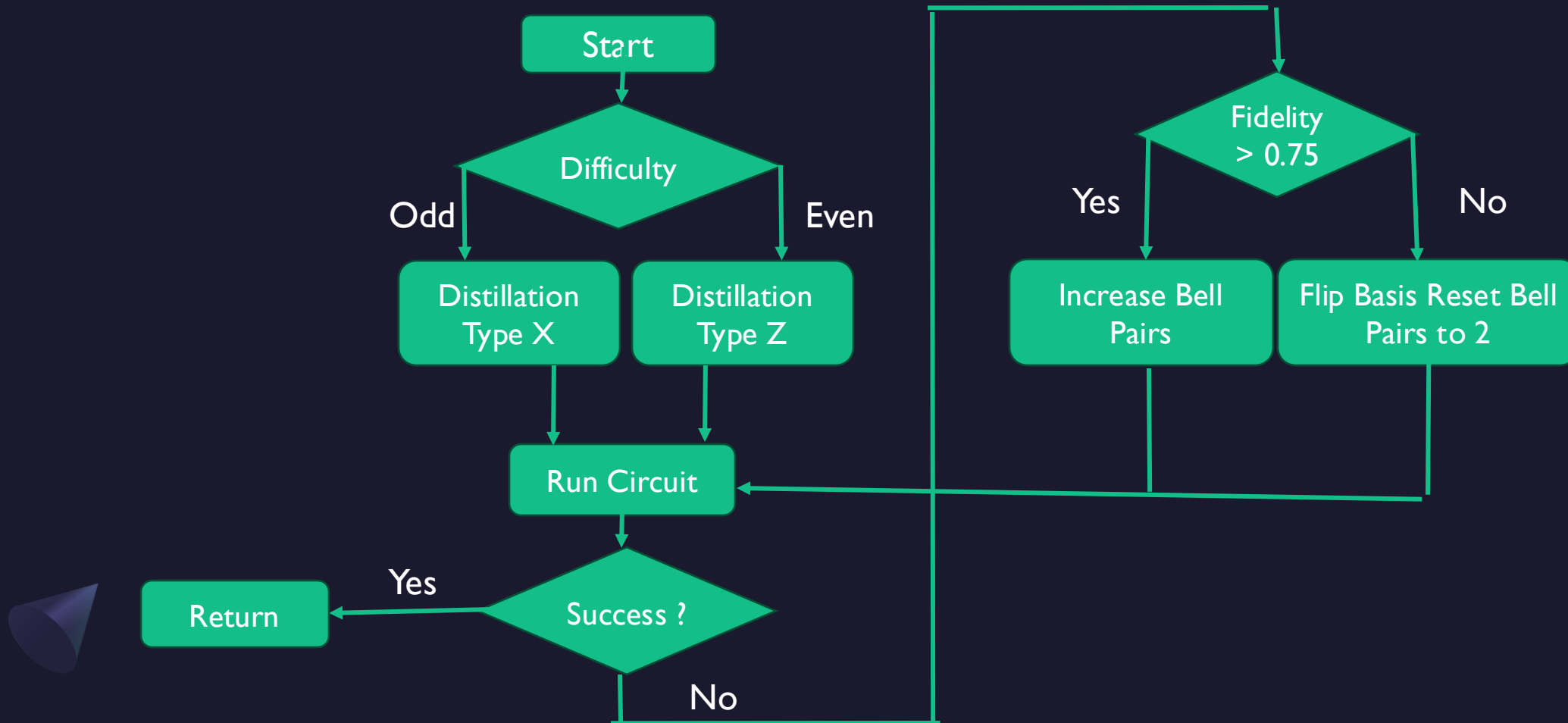
Game Strategy

Expected Value per Edge



- **Immediate node value**
 - Utility qubits
 - Bonus Bell pairs
 - Capacity (sublinear: diminishing returns)
- **Future potential**
 - Value of nearby nodes unlocked after capture (2-hop lookahead)
- **Distance bias**
 - Prefer expanding near our existing territory
- **Probability of success**
 - Learned from past attempts (bandit model)
 - Encourages safe edges early, smarter risks later

Circuit Selection Strategy



Aggressive $\leftrightarrow$ Defensive (Future Extension)

AGGRESSIVE MODE (*EXPAND FAST*)

- Focus: early dominance, map control
- Prioritizes utility qubits and high-degree nodes
- Quicker to use budget
- Accepts higher risk if expected value is high

DEFENSIVE MODE (*PLAY SAFE*)

- Focus: *long-term survival and consistency*
- Prioritizes **bonus bell pairs** and sustainability
- More **greedy resource allocation** (multiple claim attempts with incrementing Bell pairs)
- Favors safe and **high-probability captures**

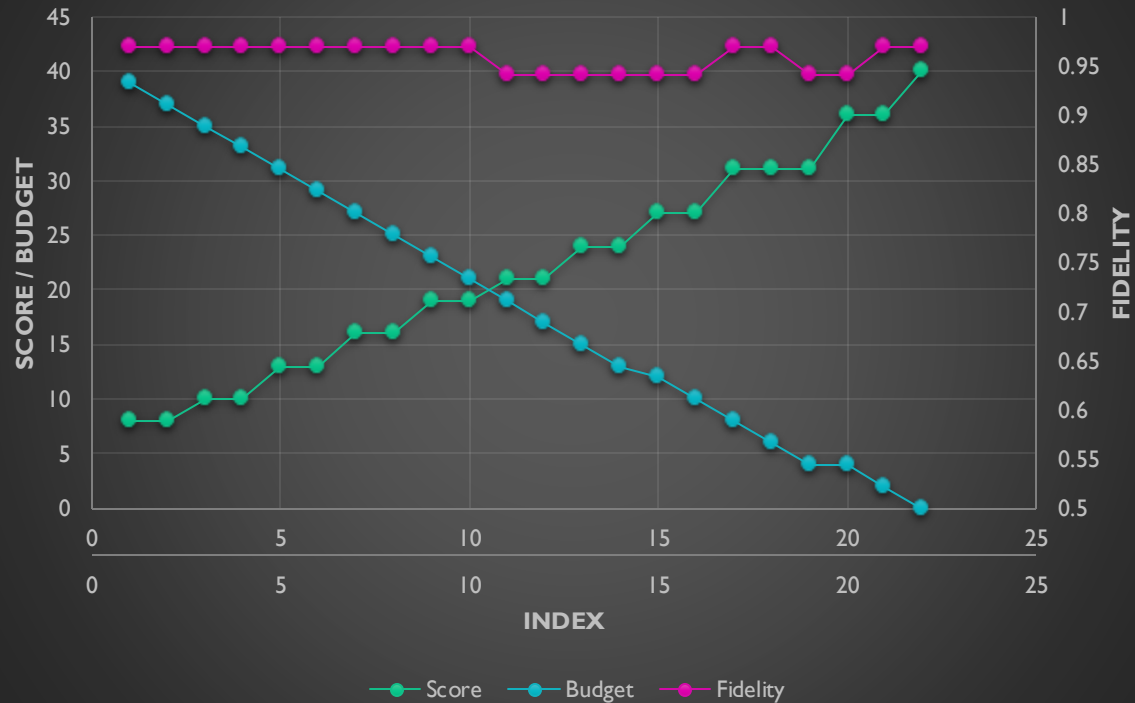




Results of Our Algorithm

Enhancing your presentation

Blizzard Agent

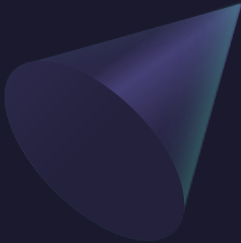


Final outcome

- A Forward-looking algorithm that :
- Probes nearby nodes
- Evaluates the best solution
- Implements a circuit tailor-made to break through!!



Future Improvements



Future Circuit Improvements

Problem	Solution
Low success probability	Employing qubit-controlled error mitigation techniques
Not quite 100% fidelity	Zero noise extrapolation
Can only check for X and Z individually	Simultaneous XZ error measurements
Exponential increase in complexity and time with number of qubits	Investigating error at the source

Network Aaron





Q&A



Citations

<https://github.com/a-aufer/qiskit/blob/master/EntanglementPurification.ipynb>

https://www.quair.group/software/pq/tutorials/locc/entanglementdistillation_bbpssw_en

[1] V. Siddhu, E. Winston, D. C. McKay, and A. Javadi-Abhari, “Basic entanglement distillation with realistic noise,” Phys. Rev. Research, vol. 7, no. 4, p. 043069, Oct. 2025, doi: 10.1103/xx9g-pkht.

